AI-Powered Conservation Intelligence for Sri Lankan Birds

Explainable Bird Habitat Risk Assessment and Conservation Recommendation

Component 04

Project ID: J26-IT-359

PROJECT PROPOSAL REPORT

Keerthihan M. - IT23189812

B.Sc. (Hons) Degree in Information Technology

Sri Lanka Institute of Information Technology

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Keerthihan	**IT23189812**	M.keerthihan

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Ms: Sanjeevi Chandrasiri	14/09/2026	Sanjeevi

ABSTRACT

Bird occurrence records can identify locations where species have been observed, but conservation decisions also require an understanding of whether those habitats are exposed to environmental threats. Habitat risk can be increased together by forest loss, tree cover loss, habitat fragmentation, land-cover change, vegetation degradation, changing weather or climate conditions and human pressure. However, these factors are spread across different biodiversity, satellite and environment datasets that are difficult for conservation users to interpret directly. This research proposes an explainable bird habitat risk assessment component by integrating bird occurrence records with multi-source environmental and geospatial data to estimate and communicate habitat risk for bird-relevant locations. At the spatial level, environmental variables will be extracted and transformed in risk indicators.

A Habitat Risk Index (HRI) will provide an interpretable risk score or category, and Random Forest and XGBoost will be explored for learning risk patterns from temporally and spatially matched historical data where a defensible target can be constructed. The candidate models will be compared using appropriate validation measures. SHAP will be used to explain environmental features contributing to the model's predictions. A rule- and evidence-based recommendation mechanism will translate the main drivers of risk into conservation actions such as habitat restoration, connectivity improvement, buffer management and increased monitoring. The component will produce an integrated environmental-risk dataset, habitat-risk map, predicted risk information, explainable risk-driver information and conservation recommendations. It provides the habitat-threat assessment and conservation-action layer of the full bird conservation intelligence framework.

Keywords: Habitat Risk Index, Habitat Threat Assessment, Random Forest, XGBoost, SHAP

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I also extend my sincere thanks to my group members Verjith, Abiththa and Sajiththiya for their cooperation, commitment, knowledge sharing and team work during the development of our research project. Their collaboration and ongoing discussions helped me a lot in integrating my habitat risk assessment component with other components of our proposed AI-powered bird conservation intelligence system.

To conclude, I would like to express my sincere thanks to my family and friends for their constant encouragement, patience, help and support during the difficult phases of this research project. Their motivation kept me on track, confident, and committed throughout my research journey in my final year.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
ML	Machine Learning
RF	Random Forest
XGBoost	Extreme Gradient Boosting
SHAP	SHapley Additive exPlanations
HRI	Habitat Risk Index
GIS	Geographic Information System
NDVI	Normalized Difference Vegetation Index
DEM	Digital Elevation Model
GBIF	Global Biodiversity Information Facility
GFC	Global Forest Change
CRS	Coordinate Reference System
AHP	Analytic Hierarchy Process

1. INTRODUCTION

1.1 Background and Context

Bird conservation is more than knowing where a bird has been seen. A place might still have bird sightings even if the environment around it is changing. The forest might be shrinking the tree cover might be getting lower the land might be breaking into pieces, the type of land cover might be changing the plants might be getting worse the rain or temperature might be different or more people might be living in the area. Because of this people who work on bird protection need a way to bring together environmental signals and find the places where birds live and where the environment might be getting worse.

The main issue that Component 04 is trying to solve is the breaking of information needed to check the health of habitats. Bird sightings are found on biodiversity websites while data about forests, land shape, weather, plants, land cover and peoples activities come from maps and sources. Looking at each of these pieces on its own makes it hard to explain the risk to the habitat and the reasons, for that risk.

This component is focused on habitat risk assessment and conservation recommendations. It does not model habitat suitability. Bird occurrence data are used to identify or link locations that're relevant to specific bird species. Environmental variables are then used to evaluate threats around these locations. Within spatial analysis units. The proposed workflow includes steps: data collection and spatial integration environmental-risk feature engineering HRI construction, machine-learning risk modeling when a reliable historical target exists, SHAP-based explanation and evidence-based conservation recommendations.

The research aims to support conservation organizations ornithologists, wildlife and forest officers, environmental authorities, researchers and protected-area managers. The work mainly aligns with Sustainable Development Goal 15. Life, on Land. Because it helps conserve biodiversity by monitoring habitat threats and guiding conservation prioritization.

Problem Definition

The research problem is that threats to bird- habitats come from many environmental factors. These factors vary in scale, units, time periods and data formats. Because of this conservation users might look at forest loss maps, weather data or bird occurrence points on their own.. They still do not have a single clear evidence-based way to see which areas are most at risk, for birds. They also don’t get an explanation of why those areas are risky or what conservation actions should be taken. Without an integrated and easy-to-understand assessment it’s hard to make decisions.

Proposed Contribution

The idea is to create a system that helps understand the risk to bird habitats. This system uses data about where birds live and the environment around them. It changes the data into easy-to-understand risk signs. Then it makes a Habitat Risk Index. It looks at how tree-based computer models can predict patterns of risk. It uses SHAP to explain the results of these models. It also gives suggestions for conservation that are supported by evidence. The new part is bringing together risk scores predicting risk explaining the results and helping with conservation actions all, in one system that focuses on birds. This new idea needs to be checked against what others have done in the literature.

1.2 Literature Review

The final literature review should critically evaluate peer-reviewed studies under four themes: ecological/habitat risk assessment; drivers of bird habitat degradation; machine learning for environmental-risk prediction; and explainable AI, for ecological decision support. Each reviewed paper should document its method, dataset, study context, findings, strengths, limitations and influence on the proposed methodology.

Tree-Based Machine Learning

Random Forest is a learning method that puts decision trees together and then mixes their results. I believe Random Forest is a choice for studying because the habitat-risk data are laid out in tables and can have complex non-straight relationships among forest, weather, terrain and human-pressure numbers. Random Forest will be used as a candidate model. Not taken for granted as the best.

XGBoost is a system that builds trees one after another and pulls them together to make predictions better. I think XGBoost is added as a candidate because boosted trees handle table data well and give a clear comparison, with Random Forest. Both Random Forest and XGBoost will be. Checked using the same test design.

Explainability

SHAP gives feature-attribution values that help explain individual model predictions and summarize how each feature contributes across a dataset [3]. In this part SHAP will address questions like whether forest loss, fragmentation, vegetation condition or human pressure had an influence, on a high predicted risk. SHAP will not be used to prove causality.

Forest Change Evidence

Global Forest Change data give detailed information, about how much tree cover existed in the past and how much forest cover has been lost each year [4]. The raw treecover2000 layer shows the amount of canopy cover at the base year, not the state of the canopy. The lossyear layer records the year when loss was first seen not the percentage of loss itself. For this reason the research will create variables. These include whether a recent loss happened what fraction of loss occurs inside a chosen area or buffer and how long it has been since the loss occurred.

Validation of Spatial Environmental Models

Spatially nearby observations often share environmental conditions and this similarity can make random train-test splitting give overly optimistic performance estimates. For this reason I prefer spatial. Grouped validation whenever the dataset allows spatial or grouped validation [5]. Spatial or grouped validation is especially relevant when risk records are generated repeatedly for geographic areas.

Table 1:- Validation of Spatial Environmental Models

Evidence	What It Supports	Research Decision
Breiman (2001) [1]	Random Forest as an ensemble tree method	Investigate RF for structured risk data.
Chen & Guestrin (2016) [2]	Regularized gradient tree boosting	Compare XGBoost with RF.
Lundberg & Lee (2017) [3]	Feature-attribution explanation	Use SHAP for model explanations.
Hansen et al. (2013) [4]	Global forest-cover change mapping	Derive forest-risk features from tree cover/loss layers.
Roberts et al. (2017) [5]	Structured/spatial validation	Avoid relying only on random splitting.

1.3 Research Gap

Existing studies have shown that machine learning techniques such as Random Forest and XGBoost work well for predicting habitat suitability and assessing quality. Explainable AI techniques such as SHAP have also been used to point out which environmental factors affect conditions. Yet most of the research mainly looks at predicting species habitat suitability or at judging overall ecosystem and landscape quality.

There is still a need for an approach that links bird occurrence information with environmental and human-caused conditions. This approach should find habitats that're both ecologically important and at risk from environmental threats.

Moreover current methods do not combine assessment clear identification of risk drivers and the creation of conservation recommendations in one decision-support system. This shortcoming matters, in the Sri Lankan context. A local explainable machine-learning framework could help focus bird-habitat conservation effectively.

Therefore this research proposes an integrated framework that brings together bird occurrence data, environmental and human-caused variables machine-learning habitat assessment, habitat risk evaluation, SHAP explainability and conservation recommendations.

2. OBJECTIVES

2.1 Main Objective

To. Test a framework that explains the risk to habitats in areas where birds live. This framework will use types of environmental information and data, about where birds are found.

2.2 Specific Objectives

1. To collect, clean and spatially integrate bird occurrence records with relevant environmental and habitat-threat datasets.
2. To design and justify environmental risk indicators and a Habitat Risk Index for bird-relevant spatial units.
3. To develop and compare Random Forest and XGBoost models for habitat-risk classification or score prediction using defensible historical risk labels.
4. To evaluate the risk models using appropriate predictive metrics and spatial or grouped validation.
5. To apply SHAP to identify the environmental variables contributing to habitat-risk predictions.
6. To generate habitat-risk maps and evidence-based conservation recommendations linked to dominant risk drivers.

3. METHODOLOGY

3.1 Research Design

The study uses a type of research that is based on numbers real-world use and experiments. Bird sightings and data, about the environment will be turned into numbers that show risk. A clear way to work with humans and robots will be. Explained. When there is past data that matches in time experiments will use machine learning to compare two methods called Random Forest and XGBoost for predicting risk. The study will check how well the models work in numbers and also check if the explanations and suggestions make sense in situations.

3.2 Study Area and Spatial Unit

The final study area must be mentioned the way in every part of the proposal. The original component document lists Northern Province as the study area. If the project has officially switched to an area such as srilanka then every filter, environmental layer, risk map and validation statement must be changed to match the new study area. I think it is important to keep the study area consistent. The analysis unit can be a grid cell, a buffer, around a bird-occurrence point or another justified spatial unit. I recommend using grid-based analysis when creating a habitat-risk map because each grid cell can receive the same set of environmental indicators.

3.3 Data Sources and Experimental Inputs

Table 2:- Data Sources and Experimental Inputs

Data Category	Fields / Layers	Role in Risk Assessment
Bird occurrence	Species, latitude, longitude, observation date, observation count and relevant effort fields	Links risk analysis to locations used by selected birds; not used as habitat-suitability output.
Tree cover	Hansen treecover2000	Baseline canopy-cover context.
Forest loss	Hansen lossyear; derived recent-loss features	Deforestation/disturbance indicator.
Weather / climate	Temperature, precipitation/rainfall and selected summaries	Environmental stress/context.
Elevation	SRTM/DEM elevation	Topographic context and possible stratification variable.
Vegetation	NDVI or equivalent vegetation indicator	Current/seasonal vegetation condition.
Land-cover change	Land Cover Change 1991-2012 shapefile	Historical land-cover transition evidence; not current land cover.
Fragmentation	Forest proportion, patch size, edge density or distance-to-edge	Measures habitat breakup/isolation.
Human pressure	Human Footprint or justified proxies such as roads/built-up/population/night lights	Anthropogenic pressure indicator.

The source PDF identifies seven core environmental features: tree cover, forest-loss year, mean temperature, annual rainfall, elevation, human footprint and fragmentation. The current implementation may add vegetation and land-cover information, but every added variable must have a clear ecological/risk justification.

3.4 Data Preparation and GIS Integration

- Filter bird occurrence records to the selected study area and species.
- Clean duplicates, invalid coordinates, unsuitable coordinate uncertainty and inconsistent dates/species names.
- Clip environmental rasters and vectors to the study boundary.
- Reproject layers to compatible coordinate reference systems.
- Resample or aggregate layers to a justified common analysis resolution.
- Extract or summarize environmental values for each grid cell or bird-relevant analysis unit.
- Convert Hansen lossyear into interpretable risk features such as recent loss, loss fraction or time since loss.
- Calculate fragmentation indicators from forest/land-cover layers.
- Create temporal summaries of temperature/rainfall that match the research question rather than using every raw weather field.
- Handle missing values and document exclusions.
- Create a final risk-analysis table with one row per spatial unit and time period.

3.5 Proposed Final Dataset Structure

Table 3:- Proposed Final Dataset Structure

Field	Example Meaning
spatial_unit_id	Unique grid/buffer identifier
species / bird_group	Bird relevance for the analysed location
latitude / longitude	Spatial reference
year / period	Time of risk observation
tree_cover	Canopy-cover indicator
recent_forest_loss	Derived loss indicator
temperature_summary	Selected temperature statistic
rainfall_summary	Selected rainfall statistic
elevation	Terrain context
human_pressure	Anthropogenic-pressure value
fragmentation	Derived habitat-fragmentation metric
ndvi	Vegetation-condition indicator
land_cover_change	Historical change/class indicator
HRI	Calculated habitat-risk score, if using HRI as labelled historical response
risk_class	Low / Moderate / High after justified thresholds

3.6 Habitat Risk Index Design

The HRI is the result from the risk assessment. Environmental variables come in units and can go in opposite directions so they need to be changed into risk indicators that can be compared. For instance more forest loss and fragmentation recently may mean risk while more tree cover might mean lower risk. Each indicator will be adjusted using a method that is clearly described and takes direction into account.

A general formula is: HRI = sum(w_i * r_i) where r_i is a normalized environmental risk indicator and w_i is its weight. The weights cannot be picked at random. They must be supported by peer-reviewed studies, input from experts with domain knowledge or a structured method like AHP (Analytic Hierarchy Process) or another approach that makes sense. It's also important to check how sensitive the results are—by doing sensitivity analysis—to see if reasonable changes in the weights lead to shifts in the final ranking of risks.

The original document shows categories such as 0–30 Low, 31–60 Moderate and 61–100 High. These numbers are examples. The actual cutoffs, for these categories should be backed up by data or expert judgment and scientific literature—not taken as universal rules that apply everywhere without testing.

3.7 Machine-Learning Target - Critical Design Decision

Machine learning should not be trained on a single static HRI that is calculated from the same predictors using a fixed formula because the model will mainly learn to copy that formula. To make the machine learning component useful the best design is risk modelling: build HRI or risk labels for many past periods using environmental data that matches each period then use conditions from an earlier period to predict risk in a later period. For example: X_t equals tree cover, recent loss, weather, fragmentation and human pressure at time t; Y_(t+1) equals HRI or risk class at the defined period. This tests whether the model learns how risk changes over time of just copying a same-time formula. If the data set does not allow a temporal target the safer scientific choice is to keep HRI as the main risk-assessment method and use Random Forest or XGBoost only for a clearly justified extra prediction task. The final report should not claim that the model can predict risk until enough past observations and a proper temporal validation design are available.

3.8 Random Forest and XGBoost

Table 4:- Random Forest and XGBoost

Model	Role	Why Included
Random Forest	Candidate habitat-risk model	Handles nonlinear structured variables and interactions; ensemble baseline.
XGBoost	Candidate habitat-risk model	Sequential regularized tree boosting; strong comparison for tabular risk data.

Both models will be trained using the training variables and evaluation partitions. Hyperparameters will be adjusted using validation data. For Random Forest parameters could include the number of trees, maximum depth minimum samples required for a split and the maximum number of features to consider. For XGBoost parameters might include the number of estimators, depth, learning rate, subsampling rate and column sampling ratio.

3.9 Baseline / Benchmark

We need a benchmark. In risk-class prediction we should use a logistic regression or a multinomial regression baseline and compare it with Random Forest and XGBoost models. For HRI prediction a simple linear regression baseline can be used. The goal is to see whether complex tree models give an improvement, over a plain statistical relationship.

3.10 Evaluation Metrics

Table 5:- Evaluation Metrics

Task	Metrics	Interpretation
Risk class prediction	Balanced Accuracy, Precision, Recall, F1-score, ROC-AUC/PR-AUC where applicable, Confusion Matrix	Measures classification quality, including imbalanced classes.
Numerical risk-score prediction	MAE, RMSE, R^2	Measures error and explained variation in continuous HRI.
Spatial/temporal generalization	Performance on held-out spatial blocks or later periods	Tests whether the model generalizes beyond nearby/training observations.

3.11 Validation Strategy

Validation will show the prediction claim. Validation will also reveal whether the component predicts risk in geographic areas and if that is the case spatially grouped validation will be used where possible.

Validation will also check if the component predicts risk from earlier environmental conditions and for that purpose temporal hold-out validation is required; for example training on earlier years and testing on later years. Validation will not rely on splitting alone, as the sole proof of generalization when spatial or temporal dependence exists.

The HRI itself also needs validation. Domain experts can check whether high-risk locations and dominant drivers are ecologically plausible. Sensitivity analysis can also test whether the ranking stays stable when reasonable changes are made to weights and thresholds.

3.12 SHAP Explainability

After selecting the best validated tree model, SHAP will be used to explain its predictions. A global SHAP summary will show which environmental variables most strongly influence risk predictions across the dataset. A local explanation will show why a particular location received a high or low predicted risk. Example interpretation: a high-risk prediction may receive positive contributions from recent forest loss, high fragmentation and high human pressure, while greater tree cover may reduce predicted risk. These are model contributions, not causal percentages.

3.13 Conservation Recommendation Engine

Dominant Risk Evidence	Example Recommendation Logic
Recent forest loss / canopy degradation	Prioritize habitat restoration, protection and monitoring in affected zones.
High fragmentation / small isolated patches	Investigate habitat connectivity, corridor or edge-management interventions.
High human pressure	Strengthen buffer-zone management, disturbance monitoring or access controls where appropriate.
Vegetation degradation	Prioritize native vegetation restoration and repeated vegetation-condition monitoring.
Climate/weather stress signal	Increase ecological monitoring and consider climate-resilient habitat management based on expert guidance.
Multiple high-risk drivers	Flag area as a priority for integrated expert review and conservation intervention.

Conservation Recommendation Engine

Recommendations will be traceable to explicit rules and supporting ecological evidence. The system will not present automated recommendations as mandatory management decisions; final conservation action remains the responsibility of qualified stakeholders.

3.14 Dynamic / Updated Risk Assessment

The prototype can handle repeated updates whenever new environmental data that are compatible come in. The processing pipeline will recalculate the indicators update the HRI and if needed run the validated model along with the SHAP explanation. As a result the risk score and recommendation shown can shift when the environmental evidence, underneath changes. This is an update mechanism, not a statement that the system reflects time ecological truth. The frequency of updates will depend on how each data source is updated and the time resolution of the data.

3.15 Development Approach

An iterative prototyping approach will be used. First data integration. Next environmental-risk feature engineering. Then HRI prototype. After that historical and temporal label construction. Next comes RF and XGBoost experiments. Then SHAP explanation. Following that recommendation logic. Then. Dashboard integration. And technical and domain validation.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

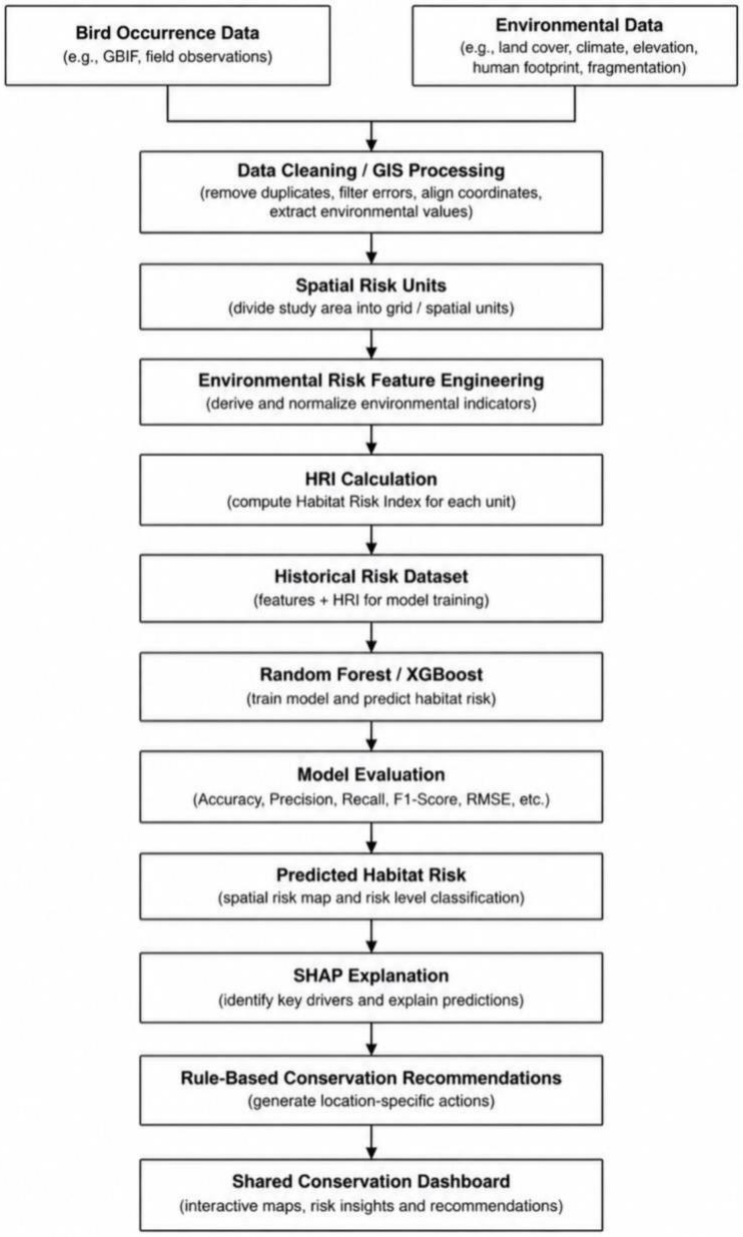


Figure 1：SYSTEM ARCHITECTURE

4.1 Integration with Overall Project

Component 04 receives bird- location/species context and environmental datasets and gives risk-focused conservation information. Component 04 does not perform image recognition, audio recognition or habitat-suitability prediction. Component 04 main outputs are habitat-risk scores/categories, risk maps, environmental-driver explanations and conservation recommendations. Component 04 outputs can be integrated into the application using shared species identifiers, geographic coordinates and agreed API/data schemas.

4.2 Individual Responsibility Boundary

Table 6: Individual Responsibility

Activity / Deliverable	My Responsibility	Shared	Other Member
Environmental dataset acquisition/integration	✓		
GIS preprocessing and risk-feature extraction	✓		
HRI methodology	✓		
Historical risk-target design	✓		
RF/XGBoost risk modelling	✓		
Risk-model validation	✓		
SHAP explanation	✓		
Recommendation rules	✓		
Risk map/output integration	✓	✓	
Common web application integration		✓	
Image/audio/other project components			Other members

5. USER REQUIREMENTS

User requirements must be supported by real elicitation evidence. The following are proposed requirements for expert/stakeholder validation and must not be described as confirmed user needs until interviews, discussions, surveys or workshops have been conducted.

5.1 Proposed Functional Requirements

FR01. The system shall allow an authorized user to select a supported bird species or bird-relevant analysis area.

FR02. The system shall display the Habitat Risk Index or validated predicted risk for mapped spatial units.

FR03. The system shall categorize risk using documented thresholds or classes.

FR04. The system shall display the environmental indicators used in the risk assessment.

FR05. The system shall show SHAP-based explanations for ML risk predictions.

FR06. The system shall identify the dominant environmental risk drivers for a selected location.

FR07. The system shall provide evidence-linked conservation recommendations based on the dominant risk drivers.

FR08. The system shall allow compatible updated environmental data to be processed so risk outputs can be recalculated.

FR09. The system shall record the model/data version associated with a displayed risk result.

5.2 Proposed Non-Functional Requirements

Table 7:Proposed Non-Functional Requirements

Category	Requirement
Usability	Risk score, category, drivers and recommendations should be understandable to conservation users without ML expertise.
Explainability	ML risk predictions should provide traceable feature-attribution information.
Reliability	The application should use validated data schemas and model versions and handle missing inputs explicitly.
Performance	A measurable response-time target will be defined after prototype benchmarking.
Maintainability	Environmental layers, thresholds and model artifacts should be replaceable through documented update procedures.
Security	Administrative data/model updates should be restricted to authorized users.
Accessibility	Risk communication should use labels/patterns in addition to color and provide readable legends.
Traceability	Each recommendation should be linked to a detected risk driver and documented rule/evidence source.

5.3 Requirements Elicitation Plan

- Ask conservation/domain experts which habitat threats are most important for selected birds.
- Ask how they currently prioritize high-risk habitat areas.
- Validate whether an HRI score and Low/Moderate/High categories are understandable and useful.
- Ask what evidence is required to trust a machine-learning risk prediction.
- Review SHAP explanation formats with stakeholders.
- Validate recommendation rules and identify actions that require expert approval.
- Document all findings and map each accepted finding to a system requirement.

6. COMMERCIALIZATION PLAN

The AI-Powered Multimodal Conservation Intelligence Framework for Sri Lankan Endemic Birds has potential for adoption as a specialized conservation technology platform. The proposed system combines image- and audio-based species identification, spatio-temporal occurrence and biodiversity prediction, and explainable habitat-threat assessment within a single framework. Its initial application is focused on Sri Lankan endemic birds, while the underlying approach may later be extended to additional species groups and conservation contexts.

6.1 Target Market and Customer Persona

The primary target market includes wildlife and conservation organizations, environmental NGOs, research institutions, universities, government environmental authorities, and biodiversity-monitoring projects. Key users include ornithologists, conservation researchers, wildlife officers, field survey teams, and environmental planners. Birdwatchers and citizen scientists form a secondary user group that may benefit from species identification and location-based observation guidance.

6.2 Value Proposition

The framework provides an integrated conservation workflow by bringing together capabilities that are commonly available through separate tools. It transforms bird images, audio recordings, occurrence records, and environmental information into interpretable outputs that can support species monitoring, field-survey planning, biodiversity assessment, habitat-risk analysis, and evidence-based conservation decision-making.

6.3 Revenue Model and Pricing Strategy

A tiered access model is proposed for future commercialization. Basic educational and citizen-science functions could remain freely accessible, while advanced analytical and institutional features could be offered through subscriptions, project-based licensing, customized deployments, or research partnerships. Additional services may include organization-specific data integration, reporting, technical support, and extended species coverage. Final pricing should be determined after operational costs and stakeholder requirements have been validated.

6.4 Cost Estimation and Competitive Advantage

The main development and operational costs are expected to arise from research effort, computing resources, data storage, system hosting, field-data collection, testing, deployment, and maintenance. Initial costs can be reduced through publicly available biodiversity and environmental datasets and open-source technologies, subject to applicable licence conditions. The framework's competitive advantage is its Sri Lankan endemic-bird focus and the integration of multimodal identification, predictive biodiversity analysis, explainable habitat assessment, and conservation-oriented outputs within one platform.

6.5 Intellectual Property Considerations

Potential intellectual property includes the original system architecture, research-generated software, trained model configurations, data-processing workflows, integration methods, and interface designs developed by the project team. Future commercialization must comply with university intellectual-property requirements and the licences or terms of use associated with third-party datasets, pretrained models, software libraries, and external services.

7. BUDGET AND JUSTIFICATION

Table 9 :BUDGET AND JUSTIFICATION

Item	Estimated Cost	Justification
Python / pandas / GeoPandas / Rasterio	Free/open source	Data processing and GIS feature extraction.
Scikit-learn / XGBoost / SHAP	Free/open source	ML and explainability.
QGIS	Free/open source	Spatial inspection, preprocessing and map validation.
Public environmental/biodiversity data	Mostly free	Core research inputs; licence/terms must be checked.
Development computer	Existing resource	Tree-based models are manageable on standard hardware for moderate datasets.
Cloud compute/storage	Optional - TBD	Needed only if local data volume or deployment requires it.
Web hosting	TBD	Prototype deployment.
Domain-expert consultation/field validation	TBD	Supports HRI, threshold and recommendation validation.

8. WORK BREAKDOWN STRUCTURE (WBS)

Table 10 :WORK BREAKDOWN STRUCTURE

Work Package	**Tasks**
WP1 Research	Problem definition, literature review, risk-method review, research gap.
WP2 Data	Bird occurrence, forest, weather/climate, elevation, vegetation, land cover, fragmentation and human-pressure data.
WP3 GIS Processing	Study-area clipping, CRS/resolution alignment, extraction, grid/buffer creation.
WP4 Risk Engineering	Normalize indicators, derive forest-loss/fragmentation variables, design HRI and thresholds.
WP5 ML Experiments	Construct historical labels, baseline, Random Forest, XGBoost, tuning.
WP6 Evaluation	Spatial/temporal validation, classification/regression metrics, sensitivity analysis.
WP7 Explainability	Global/local SHAP analysis.
WP8 Recommendations	Evidence-based rule design and expert review.
WP9 Product	Risk map, dashboard integration, update workflow.
WP10 Documentation	Analysis, proposal/thesis writing, final evidence and decision log.

9. GANTT CHART



Figure 2:Gantt Chart

10. REFERENCES

[[1] L. Breiman, “Random Forests,” *Machine Learning*, vol. 45, pp. 5–32, 2001, doi: 10.1023/A:1010933404324.

[2] T. Chen and C. Guestrin, “XGBoost: A Scalable Tree Boosting System,” in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2016, pp. 785–794, doi: 10.1145/2939672.2939785.

[3] S. M. Lundberg and S.-I. Lee, “A Unified Approach to Interpreting Model Predictions,” in *Advances in Neural Information Processing Systems 30*, 2017.

[4] M. C. Hansen *et al*., “High-Resolution Global Maps of 21st-Century Forest Cover Change,” *Science*, vol. 342, no. 6160, pp. 850–853, 2013, doi: 10.1126/science.1244693.

[5] D. R. Roberts *et al*., “Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure,” *Ecography*, vol. 40, no. 8, pp. 913–929, 2017, doi: 10.1111/ecog.02881.

[6] S. Liu and J. Huang, “Spatio-Temporal Evolution and Driving Forces of Habitat Quality in China’s Arid and Semi-Arid Regions: An Interpretable Machine Learning Perspective for Ecological Management,” *Land*, vol. 14, no. 10, Art. no. 1937, 2025, doi: 10.3390/land14101937.

land-14-01937.pdf

[7] T. B. Gelete, D. Tulu, K. H. Yasin, and E. Kebede, “Machine learning predictions of climate change effects on nearly threatened bird species (*Crithagra xantholaema*) habitat in Ethiopia for conservation strategies,” *Scientific Reports*, vol. 15, Art. no. 36972, 2025, doi: 10.1038/s41598-025-20952-4.

s41598-025-20952-4 (4).pdf

[8] Z. Zhao, Y. Lei, X. Jiang, Y. Leng, W. Li, and W. Zhao, “Explainable machine learning reveals drivers of ecological quality in a karst watershed: implications for sustainable management of the Mawei River Basin, China,” *International Journal of Digital Earth*, vol. 19, no. 1, Art. no. 2600876, 2026, doi: 10.1080/17538947.2025.2600876.

FINAL REFERENCE REQUIREMENT: Add and verify peer-reviewed sources specifically supporting bird-habitat threats, fragmentation, vegetation/NDVI, human pressure, HRI or ecological-risk weighting, conservation recommendation rules, Sri Lankan conservation context and the exact datasets used. Every important claim and methodological decision must be evidence-backed.

11. APPENDICES

Appendix A - Preliminary Risk Dataset

Table 11:-Appendix A

Variable Group	Candidate Fields
Bird	common_name, scientific_name, latitude, longitude, observation_date, observation_count, protocol/effort fields
Weather	temperature_mean/max/min, precipitation_sum, rain_sum, windspeed_max; reduced to justified summaries
Terrain	elevation
Forest	treecover2000, lossyear-derived features
Vegetation	NDVI/current vegetation indicator
Land cover	Land Cover Change 1991-2012 and/or newer current land-cover layer
Derived risk	fragmentation, human pressure, HRI, risk_class

Appendix B - Stakeholder / Domain Expert Questions

7. Which environmental factors most strongly indicate bird-habitat degradation in the study area?
8. How should habitat risk be communicated: numeric score, categories, map, or a combination?
9. What evidence would make a high-risk alert credible to you?
10. Which HRI weighting or threshold decisions require expert validation?
11. What conservation actions are appropriate for forest loss, fragmentation, vegetation degradation and human pressure?
12. How frequently would updated risk information be useful?
13. Which recommendations should never be automated without expert review?

Appendix C - Mandatory Proposal Evidence and Decision Log

Table 12 :Appendix c

Date	Claim / Problem	Evidence Collected	Alternatives Considered	Decision Made	Supervisor Discussion
2nd August 2026	The research team discussed whether the project should focus on bird species worldwide or Sri Lankan endemic bird species.	The team considered a worldwide bird scope and a Sri Lankan endemic-bird-specific scope. The research focus, feasibility, conservation relevance, and project scope were discussed with the supervisor.	Worldwide bird species; Sri Lankan bird species; Sri Lankan endemic bird species.	The research was scoped specifically to Sri Lankan endemic bird species to maintain a focused, locally relevant, and manageable research scope.	The alternatives were discussed with the supervisor, and the project focus was refined to Sri Lankan endemic bird species.
15th August 2026	The research team needed to determine whether the externally obtained bird dataset was suitable for the proposed project.	The externally obtained dataset was presented to the supervisor and reviewed against the project's selected species scope and research requirements.	Use the external dataset; rely only on publicly available datasets; combine external and appropriate public data sources where required.	The external dataset was accepted as supporting data for the project, with each component preparing and preprocessing the relevant data according to its individual research requirements.	The dataset was presented and discussed with the supervisor, including its relevance and suitability for the proposed research components.

Appendix D - AI Use Disclosure

Table 13:-Appendix A

AI Tool	Purpose	Generated / Assisted	Verification	Final Student Contribution
ChatGPT	Proposal structuring and methodological explanation	Draft wording, organization and clarification	Student verifies against source PDF, peer-reviewed research, dataset	Student makes research decisions, validates data/methodology,
			documentation and supervisor/domain feedback	conducts experiments, revises and defends final work

No AI-generated technical diagram should be submitted. Draw the system architecture manually in black and white as required by the proposal template.

Appendix E :- External Supervisor Dataset Approval and Dataset Evidence

The project has received dataset approval and supporting dataset material from the external supervisor. The approved material will be stored in the project Google Drive and provided as supporting evidence for the proposed research. This allows the approval and dataset evidence to be reviewed without embedding large data files directly in the proposal.

Dataset Approval and Dataset Evidence: Research_Dataset External approval Letter

Appendix F : Prototype Evidence

The project prototype provides supporting evidence of the proposed system design and demonstrates how the main research components and user-facing outputs are intended to be presented through the integrated application.

Prototype Link: System prototype

12. CONCLUSION

In this work we introduce the Explainable Habitat Risk Assessment and Conservation Recommendation component for the AI-Powered Conservation Intelligence System for Sri Lankan Birds. This component allows the system to evaluate bird habitat risk by combining bird occurrence data with environmental factors like tree cover, forest loss, temperature, rainfall, elevation, human pressure, habitat fragmentation, vegetation condition and land-cover change.

Within the system environmental data and bird occurrence data are first cleaned and combined using GIS methods. The environmental variables are then converted into risk-related features. Used to calculate the Habitat Risk Index (HRI). Random Forest and XGBoost models will be. Compared to predict habitat risk using the historical risk data, which helps the system find areas that have low, moderate or high habitat risk.

The Explainable AI component using SHAP will offer explanations for the predicted habitat risk by showing which environmental factors have the biggest impact on each prediction. Based on the risk level and main environmental threats a rule-based conservation recommendation system will suggest actions like habitat restoration, better habitat connectivity, less human disturbance and focusing on high-risk areas for monitoring.

The main contribution of Component 04 is an integrated, explainable and data-driven way to assess habitat risk. This moves beyond showing environmental datasets by offering a combined risk assessment, clear reasons for the main risk factors and real steps for conservation. The component is also made to work with environmental data so the risk assessment and recommendations can be updated as conditions change.

The component will be tested by comparing Random Forest and XGBoost using classification or regression measures checking performance on new data looking at how consistent and clear the SHAP explanations are, doing sensitivity analysis on the HRI where needed and getting feedback from experts, on the risk factors and conservation advice.